

Trenton Pham

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EDUCATION

University of California, San Diego

La Jolla, CA

Bachelor of Science, Data Science | GPA: 3.7/4.0 (Provost Honors)

Expected Graduation, June 2028

- Relevant Coursework: Data Structures, Theoretical Foundations of Data Science, Linear Algebra
- Organizations: Institute of Electrical & Electronics Engineers, Association for Computing Machinery, Club Swim

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn), Java, SQL, MATLAB

Developer Tools: Git, GitHub, Jupyter Notebook, MySQL

Data Modeling & Statistics: Data Wrangling, Data Visualization, Hypothesis Testing, Significance Testing

PROJECTS

Post-secondary Academic Performance Predictive Modeling

February 2026 – Present

- Co-designed and documented a predictive-modeling study of pre-college indicators using a 23,000-row, 9,000-column longitudinal dataset (HLS 2009-16).
- Engineered data pipeline that trimmed the raw 20,000-student file to 1,200 usable variables, removed 80% of missing/constant columns, and produced a clean 7,000-row analytic table.
- Performed EDA on 30+ variables, generating 15+ plots (histograms, boxplots, scatterplots, pairplots) and deriving 5 new features to explore correlations.

Sitcom Analysis

November 2025 – December 2025

- Analyzed datasets of the American sitcom, *Friends* (1994), with Pandas to extract insights from 236 rows of TV episodes, dialogue, and sentiment data.
- Developed 8 reusable functions for data cleaning, aggregation, bootstrapping, and hypothesis testing.
- Used bootstrapping and 99% confidence intervals to visualize uncertainty with Matplotlib and make statistically supported recommendations based on ratings, viewership, and gender balance.

Music Analysis & Recommendation

October 2025

- Conducted exploratory data analysis on 230+ Taylor Swift songs across 11 albums using Pandas, computing grouped statistics and visualizing trends in popularity, audio features, and release year.
- Implemented a content-based recommendation system comparing 7 normalized audio features using Euclidean distance to identify musically similar tracks.

WORK & VOLUNTEER EXPERIENCE

Lifeguard & Swim Instructor

May 2023 – Present

Seattle Parks and Recreation

Seattle, WA

- Revitalized a late-night teen program that provided free lifeguard training in a low-income, diverse region of Seattle, resulting in 5+ lifeguard hires within Seattle and maintaining an attendance of 15+ teens in the 2025-2026 cycle.
- Taught lessons to 30+ youths and adults in skills ranging from water safety to pre-competitive swimming, leading to positive feedback and further quarterly enrollment.

Swim Meet Data Manager

September 2024 – November 2024

Liberty High School

Issaquah, WA

- Developed proficiency in the meet manager software and organized time sheets, resulting in competition results with 100+ manual data entries being published up to 48 hours earlier compared to previous seasons.